

Program 3 : Build a production-ready Docker container for a Node.js application.

File structure:

/app.js

/package.json

/Dockerfile

Steps to create and run the application:

1. Initialize the Node.js project and install Express:

```
$ npm init -y
```

```
$ npm install express
```

2. In package.json, ensure the scripts section includes:

```
"scripts": {  
  "start": "node app.js"  
}
```

3. Create app.js file with the following code:

```
-----  
const express = require("express");  
const app = express();  
  
app.get("/", (req, res) => {  
  res.send("doing 3rd program");  
});  
  
const PORT = 8000;  
app.listen(PORT, "0.0.0.0", () => {  
  console.log(`Server running on http://localhost:${PORT}`);  
});  
-----
```

4. Create a Dockerfile with the following content:

```
-----  
FROM node:18-alpine  
WORKDIR /app  
COPY package*.json ./  
RUN npm install --production  
COPY . .  
EXPOSE 8000  
CMD ["npm", "start"]  
-----
```

5. To build the Docker image:

```
$ docker build -t program3 .
```

6. To list available images:

```
$ docker images
```

7. To run the container:

```
$ docker run -d -p 8000:8000 program3
```

8. To see running containers:

```
$ docker ps
```

9. To test the application output:

```
$ curl http://localhost:8000
```

Execution Output:

```
girish-yandigeri@girish-yandigeri-Lenovo-Legion-Y540-15IRH-PG0: ~/devops/normallabs/lab3$ ls  
app.js  Dockerfile  package.json  
girish-yandigeri@girish-yandigeri-Lenovo-Legion-Y540-15IRH-PG0: ~/devops/normallabs/lab3$ docker build -t program3 .  
[+] Building 3.4s (10/10) FINISHED  
=> [Internal] load build definition from Dockerfile 0.3s  
=> => transferring dockerfile: 163B 0.0s  
=> [Internal] load metadata for docker.io/library/node:18-alpine 1.3s  
=> [Internal] load .dockerignore 0.2s  
=> => transferring context: 2B 0.0s  
=> [1/5] FROM docker.io/library/node:18-alpine@sha256:8d6421d663b4c28fd3ebc498332f249011d118945588d0a35cb9bc4b8ca09d9e 0.0s  
=> [Internal] load build context 0.2s  
=> => transferring context: 89B 0.0s  
=> CACHED [2/5] WORKDIR /app 0.0s  
=> CACHED [3/5] COPY package*.json ./ 0.0s  
=> CACHED [4/5] RUN npm install --production 0.0s  
=> CACHED [5/5] COPY . . 0.0s  
=> exporting to image 0.2s  
=> => exporting layers 0.0s  
=> => writing image sha256:8866b884a4a8695fa0929cbce9eef8930e35683a5b32d972d2d8372760d336 0.0s  
=> => naming to docker.io/library/program3 0.0s  
7b52d9ca1c658c318471cd59f248f69ca5529392c4b7f1e0a54fc987293b1af  
girish-yandigeri@girish-yandigeri-Lenovo-Legion-Y540-15IRH-PG0: ~/devops/normallabs/lab3$ docker run -d -p 8000:8000 program3  
girish-yandigeri@girish-yandigeri-Lenovo-Legion-Y540-15IRH-PG0: ~/devops/normallabs/lab3$ docker ps  
CONTAINER ID   IMAGE          COMMAND                  CREATED        STATUS        PORTS                               NAMES  
7b52d9ca1c65   program3      "docker-entrypoint.s..." 5 seconds ago Up 3 seconds  0.0.0.0:8000->8000/tcp, [::]:8000->8000/tcp  sweet_mcclintock  
c83b2fdaf18c   program2      "docker-entrypoint.s..." 11 minutes ago Up 11 minutes  0.0.0.0:3000->3000/tcp, [::]:3000->3000/tcp  gallant_galois  
38c67a115ddf   program1      "docker-entrypoint.s..." 20 minutes ago Up 20 minutes  0.0.0.0:4000->4000/tcp, [::]:4000->4000/tcp, 5000/tcp  mystifying_tharp  
girish-yandigeri@girish-yandigeri-Lenovo-Legion-Y540-15IRH-PG0: ~/devops/normallabs/lab3$ curl http://localhost:8000  
doing 3rd programgirish-yandigeri@girish-yandigeri-Lenovo-Legion-Y540-15IRH-PG0: ~/devops/normallabs/lab3$
```